

## projects do you expect to offer these commercially?

We expect three to four companies to be commercially offering both BGA and flip-chip packaging by then. By 2027, India should have multiple domestic companies offering commercial BGA and flip-chip BGA packaging, with more advanced packaging scaling up from 2028 onwards.

## The DLI scheme has created considerable momentum for India's fabless ecosystem. How do you see it evolving under Semicon India 2.0?

The DLI scheme operates through three levels of support.

The first is access to world-class electronic design automation (EDA) tools, not only for startups and micro, small, and medium enterprises (MSMEs) but also for academic institutions, enabling teaching, research, and commercial product development without high upfront costs.

The second is seed funding, which has enabled 24 approved startups to begin designing proprietary chips. Their technologies have attracted institutional venture capital because investors recognise their commercial potential. Collectively, these companies have attracted VC funding exceeding four times the total incentives disbursed under the DLI scheme, demonstrating the scheme's effectiveness in catalysing private investment and strengthening the domestic semiconductor design ecosystem.

The third is the deployment-linked incentive. Although none of the startups has yet reached this stage, we expect the first commercial chip deployments within the next two to three months. Around a year after successful market deployment, these companies will become eligible for deployment-linked incentives.

Our experience has also highlighted structural gaps that will be

addressed under Semicon India 2.0. Even so, the current policy has successfully encouraged 24 hardware startups to enter this highly challenging sector.

**The next phase will significantly strengthen India's fabless ecosystem by extending participation beyond startups and MSMEs to large Indian companies.** We are also introducing specialised financial and operational support mechanisms. Although I cannot disclose the details until the policy receives final financial approval, it will provide a comprehensive framework for scaling India's fabless semiconductor industry.

## India currently consumes around US\$50 billion worth of semiconductors annually. Given this opportunity, what is a realistic revenue target for India's fabless sector by 2030?

The key challenge for Indian fabless startups is competing with two established groups: western multinational companies and Chinese semiconductor firms.

Displacing Chinese suppliers will depend significantly on government policy. Many Chinese companies benefit from opaque cost structures, while India also has cybersecurity and national security concerns around imported hardware. Accordingly, the government is introducing trusted sourcing regulations. A good example is MeitY's recent framework for CCTV and public surveillance systems, which encourages electronics manufacturing services (EMS) companies to adopt trusted, locally designed silicon. Such measures will create a stable domestic market for Indian fabless companies.

Against Western competitors, I believe Indian firms can compete effectively on cost because of government support, India's engineering talent, and lower design costs.

Rather than setting a rigid revenue target, it is important to recognise that startups are

inherently unpredictable; many will fail, some will be acquired, and a few will grow into major companies. Nevertheless, based on the overall trajectory, **I believe Indian fabless companies can collectively achieve US\$2 billion-US\$3 billion in revenue by 2030, with the first commercial tape-outs already taking place in 2026, followed by successive waves in 2027 and 2028.** As these companies build customer confidence, we expect India to develop a mature fabless ecosystem with 10-15 globally significant design companies.

## As a conservative benchmark for the ISM, would it be reasonable to target aggregate revenues of US\$100-500 million for Indian fabless companies by 2030?

Yes. That is a very reasonable and conservative baseline. I believe the industry will significantly exceed that range. The operational issues affecting this growth have already been discussed extensively within the ISM, and we are prepared to address them structurally under Semicon India 2.0.

## Chip design is a major focus of the DLI scheme. The Chips to Startup (C2S) programme has become one of the world's largest academic semiconductor initiatives. Could you outline the programme's structure and achievements so far?

The C2S programme has been highly successful. We initially set a target of training 85,000 semiconductor engineers and have already upskilled 67,000.

Under Semicon India 2.0, the programme will expand further, with participation expected to reach at least 500 institutions. Beyond providing EDA tools, C2S enables students to take their designs through fabrication and packaging, giving them hands-on experience with physical silicon and preparing them as industry-ready chip designers.